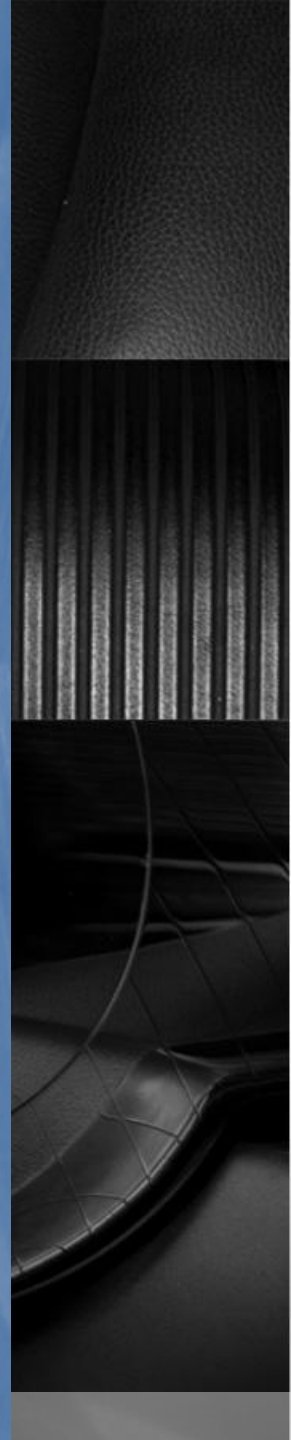
COMP4431 Artificial Intelligence

Topics on Probability (Supp.)

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Conditional probability? Joint probability?

- What is the difference between

  $P(A|B)$ and $P(A)$ and $P(A \cap B)$?

Weather and Traffic



Event A: “There is a traffic jam on the way to campus.”

Event B: “It is raining.”



- Is the chance of a traffic jam the same on a sunny day as on a rainy day?

Weather and Traffic

- How about if you are told it’s raining, do you revise how likely you think a traffic jam is?

Weather and Traffic

	Traffic jam (A)	No jam ($\overline{A}$)	Total
Raining (B)	20	20	40
Not raining ($\overline{B}$)	10	50	60
Total	30	70	100

1.P(traffic jam)?
P(A) = 30/100 = 0.3

2. P(traffic jam | it is raining)?
P(A | B) = 20/40 = 0.5

3. P(traffic jam | it is not raining)?
P(A | $\overline{B}$) = 10/60 ≈ 0.167

4. P(traffic jam ∩ it is raining)?
P(A ∩ B) = 20/100 = 0.2

Use Bayes Theorem :

P(A | B) = P(A ∩ B) / P(B)
= 0.2 / 0.4
= 0.5

Law of Total Probability

- $P(A) = P(A \cap B) + P(A \cap \overline{B}) \qquad = \sum_B P(A \cap B)$

	Traffic jam (A)	No jam ($\overline{A}$)	Total
Raining (B)	20	20	40
Not raining ($\overline{B}$)	10	50	60
Total	30	70	100

- P(A) = 20/100 + 10/100 = 0.2+0.1

  = 0.3

Throwing 2 Dice

- Finally, a simple question
- Event D is throwing a fair die with 6 as result
- Then, how to notate throwing 2 dice with both 6? And how to calculate the result?
- What assumption you had made?